

```
import cv2
import numpy as np
import matplotlib.pyplot as plt

# Create a black image
img = np.zeros((500, 700, 3), dtype=np.uint8)

# Draw shapes
cv2.rectangle(img, (80, 100), (250, 270), (255, 255, 255), 3)

cv2.line(img, (300, 100), (550, 350), (255, 255, 255), 3)
cv2.line(img, (550, 100), (300, 350), (255, 255, 255), 3)

cv2.line(img, (100, 400), (600, 400), (255, 255, 255), 3)

# Convert to grayscale
gray = cv2.cvtColor(img, cv2.COLOR_BGR2GRAY)

# Detect edges
edges = cv2.Canny(gray, 50, 150)

# Hough Transform parameter settings
settings = [
    (150, 100, 10),
    (100, 80, 20),
    (50, 50, 30),
    (200, 120, 5)
]

plt.figure(figsize=(15, 9))

# Original image
plt.subplot(2, 3, 1)
plt.imshow(cv2.cvtColor(img, cv2.COLOR_BGR2RGB))
plt.title("Original Image")
plt.axis("off")
```

```
# Canny edges
plt.subplot(2, 3, 2)
plt.imshow(edges, cmap="gray")
plt.title("Canny Edges")
plt.axis("off")

# Apply Hough Transform
for i, (threshold, minLineLength, maxLineGap) in enumerate(settings):

    lines = cv2.HoughLinesP(
        edges,
        rho=1,
        theta=np.pi / 180,
        threshold=threshold,
        minLineLength=minLineLength,
        maxLineGap=maxLineGap
    )

    result = img.copy()

    count = 0

    if lines is not None:

        # Convert lines into 4-column format
        lines = np.asarray(lines).reshape(-1, 4)

        count = len(lines)

        for x1, y1, x2, y2 in lines:

            cv2.line(
                result,
                (int(x1), int(y1)),
                (int(x2), int(y2)),
                (0, 0, 255),
                2
            )
```

```
# Display result
plt.subplot(2, 3, i + 3)

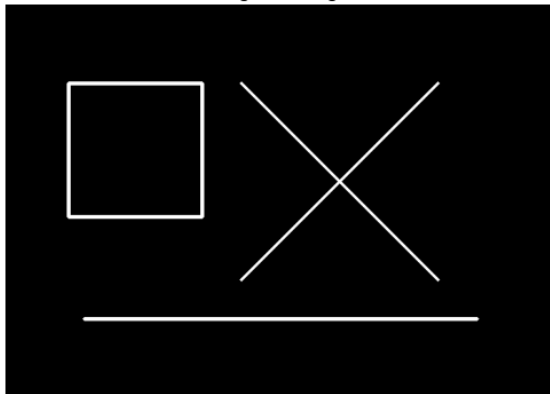
plt.imshow(
    cv2.cvtColor(result, cv2.COLOR_BGR2RGB)
)

plt.title(
    f"Setting {i + 1}\n"
    f"Threshold={threshold}, Lines={count}"
)

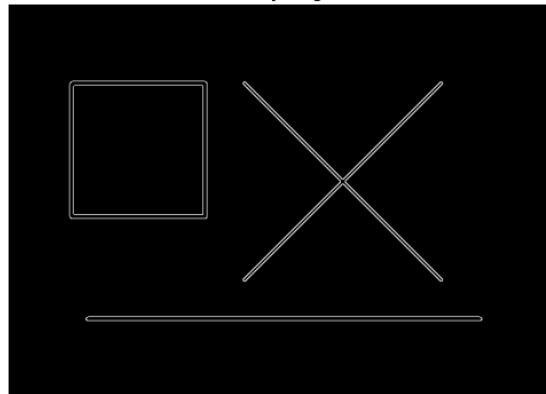
plt.axis("off")

plt.tight_layout()
plt.show()
```

Original Image



Canny Edges

Setting 1
Threshold=150, Lines=18Setting 2
Threshold=100, Lines=18Setting 3
Threshold=50, Lines=18Setting 4
Threshold=200, Lines=10

